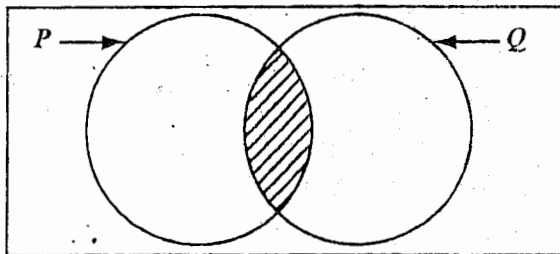


1. $(-3)^2 + (-2)^2 =$
- (A) -13
(B) -10
(C) 13
(D) 25
2. $11.1 \div 0.01$
- (A) 110
(B) 111
(C) 1 100
(D) 1 110
3. The number 3076 written in standard form is
- (A) 3.076×10^3
(B) 3.076×10^2
(C) 3.076×10^{-2}
(D) 3.076×10^{-3}
4. If \$560 is shared in the ratio 2 : 3 : 9, the difference between the largest and the smallest shares is
- (A) \$ 80
(B) \$240
(C) \$280
(D) \$360
5. In a school the ratio of the number of pupils to the number of teachers is 20 : 1. If the number of pupils is 840, how many teachers are there?
- (A) 40
(B) 42
(C) 800
(D) 840
6. There are 40 students in a class. Girls make up 60% of the class. 25% of the girls wear glasses. How many girls in the class wear glasses?
- (A) 6
(B) 8
(C) 10
(D) 15
7. A student used a rod to measure the distances 3.6 m, 5.4 m and 7.2 m. What was the GREATEST length of the rod if it fitted each distance an exact number of times?
- (A) 0.9 m
(B) 1.8 m
(C) 2.7 m
(D) 3.6 m
8. 301 can be written as
- (A) $3 \times 10^3 + 1 \times 10$
(B) $3 \times 10^2 + 1 \times 10$
(C) $3 \times 10^3 + 1$
(D) $3 \times 10^2 + 1$
9. $-\left(\frac{1}{2}\right)^3$ is the same as
- (A) $-\frac{1}{8}$
(B) $-\frac{1}{6}$
(C) $\frac{1}{8}$
(D) $\frac{1}{6}$
10. By the distributive law $49 \times 17 + 3 \times 49 =$
- (A) $52 + 66$
(B) 52×66
(C) $49 + 20$
(D) 49×20

11. If $P = \{a, b, c\}$ then the number of subsets of P is

(A) 3
(B) 4
(C) 8
(D) 16

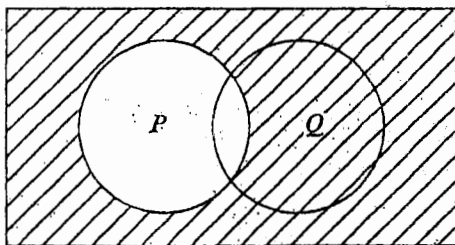
Item 12 refers to the Venn diagram below.



12. The two circles above represent set P and set Q . If $P = \{\text{Factors of 6}\}$ and $Q = \{\text{Factors of 4}\}$, then the shaded region represents

(A) $\{ \}$
(B) $\{1, 2\}$
(C) $\{4, 6, 8, \dots\}$
(D) $\{12, 24, 36, \dots\}$

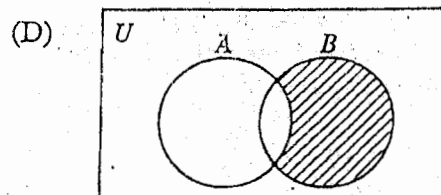
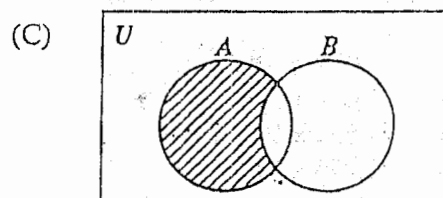
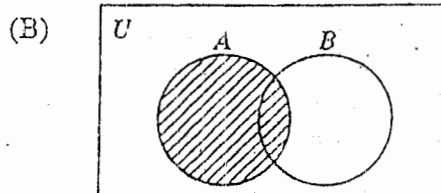
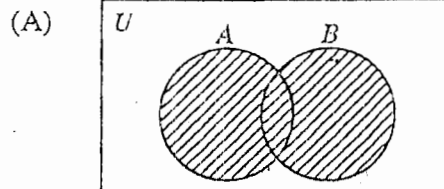
Item 13 refers to the Venn diagram below.



13. In the Venn diagram above, the shaded area represents

(A) P'
(B) $(P \cup Q)'$
(C) $Q \cap P'$
(D) $Q \cup P'$

14. In which one of the following Venn diagrams is the region $A \cap B'$ shaded?



15. Tom bought a pen for \$60 and sold it to gain 20% on his cost price. How much money did he gain?

(A) \$12
(B) \$40
(C) \$72
(D) \$80

16. An article costs \$161. If a profit of 13% is to be made the selling price, in dollars, is

(A) $161(1 + \frac{13}{100})$
(B) $13(1 + \frac{161}{100})$
(C) $161(13 + \frac{13}{100})$
(D) $13(161 + \frac{13}{100})$

17. Mr. Duncan bought a table at a discount of 30% thus saving \$42. What was the marked price of the table?

(A) \$ 98
(B) \$110
(C) \$140
(D) \$182

18. A television set costs \$350 cash. When bought on hire-purchase, a deposit of \$35 is required, followed by 12 monthly payments of \$30. How much is saved by paying cash?

(A) \$10
(B) \$25
(C) \$40
(D) \$45

19. A company employs 12 gardeners at \$26 per day, and 8 clerks at \$17 per day. What is the mean daily wage, in dollars, of the 20 employees?

(A) \$20.00
(B) \$21.50
(C) \$22.40
(D) \$31.50

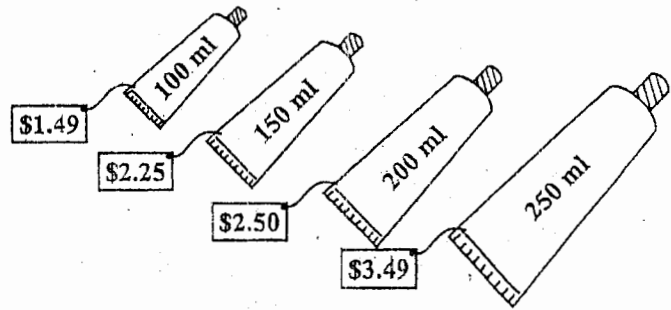
20. If \$7000 is borrowed at the rate of 5% per annum for 3 years, the simple interest is

(A) \$ 105
(B) \$ 210
(C) \$ 370
(D) \$1050

21. A man's taxable income is found to be \$15 200. He pays tax at the rate of 25%. The amount of income tax he pays is

(A) \$3 775
(B) \$3 800
(C) \$3 825
(D) \$3 875

Item 22 refers to toothpaste which is sold in four different tubes.



22. Which amount is the best buy?

(A) 100 ml
(B) 150 ml
(C) 200 ml
(D) 250 ml

23. $2x - 3(x + 8) =$

(A) $x + 8$
(B) $-x - 24$
(C) $5x + 8$
(D) $-5x - 24$

24. $5(x + y) - 3(x - y) =$

(A) $2x$
(B) $2x + 2y$
(C) $2x + 8y$
(D) $8x + 8y$

25. For all a and b , $3a(a + 2b) - b(2a - 3b) =$

(A) $3a^2 - ab + 3b^2$
(B) $3a^2 + 4ab + 3b^2$
(C) $3a^2 + 4ab - 3b^2$
(D) $3a^2 + 8ab - 3b^2$

26. If $a * b = \frac{b}{a} - 1$, then $7 * 28 =$

(A) $-\frac{3}{4}$

(B) $\frac{1}{4}$

(C) 3

(D) 4

27. When simplified, $\frac{4x}{7y} + \frac{3x}{5y}$ is written as

(A) $\frac{41x}{35y}$

(B) $\frac{41x^2}{35y}$

(C) $\frac{41xy}{35y}$

(D) $\frac{20x + 21y}{35y}$

28. $3^x \times 3^y =$

(A) 9^{xy}

(B) 3^{xy}

(C) 9^{x+y}

(D) 3^{x+y}

29. If $20a - 16 = 12(3 - a)$, then $a =$

(A) $\frac{5}{8}$

(B) $\frac{13}{8}$

(C) $\frac{21}{16}$

(D) $\frac{5}{2}$

30. If x is an integer which satisfies the inequalities $4 < x - 2 < 8$, then the smallest possible value of x is

(A) 4

(B) 5

(C) 6

(D) 7

31. The sum of two numbers, x and y , is 18, and their difference is 14. Which pair of equations below describes the above statement?

(A) $2(x + y) = 18$

$2(x - y) = 4$

(B) $2(xy) = 18$

$2(x - y) = 4$

(C) $(x + y) = 18$

$(x - y) = 14$

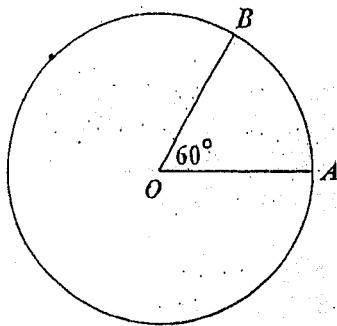
(D) $(x + y) = 22$

$(x - y) = 14$

32. The distance around the edge of a circular pond is 88 m. The radius, in metres is

- (A) 88π
 (B) 176π
 (C) $\frac{88}{\pi}$
 (D) $\frac{88}{2\pi}$

Item 33 refers to the circle below with centre O and circumference of 20 cm.



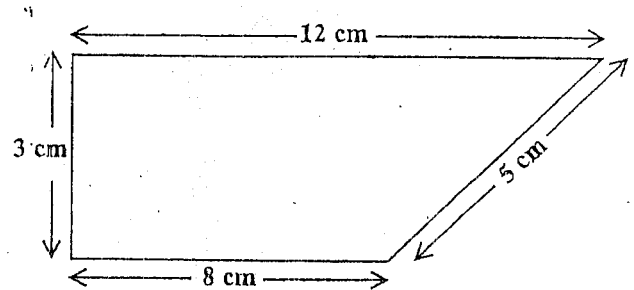
33. The length of the minor arc AB, in cm, is

- (A) $\frac{1}{60} \times 20$
 (B) $\frac{60}{360} \times 20$
 (C) $\left(\frac{360-60}{360}\right) \times 20$
 (D) 60×20

34. If the length of a rectangle is doubled, by what number must the width be multiplied in order that the area remains the same?

- (A) 3
 (B) 2
 (C) $\frac{1}{2}$
 (D) $\frac{1}{4}$

Item 35 refers to the trapezium below.



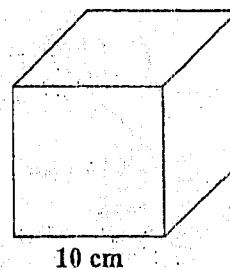
35. The area of the trapezium above is

- (A) 24 cm^2
 (B) 28 cm^2
 (C) 30 cm^2
 (D) 36 cm^2

36. The area of a rectangle is 53.6 cm^2 . If the length is multiplied by four and the width is halved, the area would then be

- (A) 214.4 cm^2
 (B) 107.2 cm^2
 (C) 53.6 cm^2
 (D) 26.8 cm^2

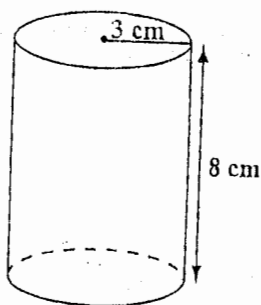
Item 37 refers to the cube below.



37. The volume of the cube is

- (A) $1\,000 \text{ cm}^3$
 (B) 300 cm^3
 (C) 100 cm^3
 (D) 30 cm^3

Item 38 refers to the cylinder below with radius 3 cm and height 8 cm.



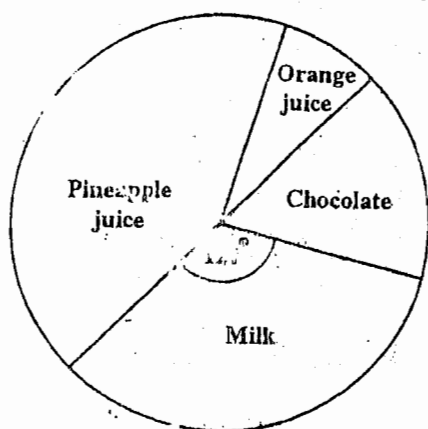
38. The volume of the cylinder is

- (A) $12\pi \text{ cm}^3$
- (B) $48\pi \text{ cm}^3$
- (C) $72\pi \text{ cm}^3$
- (D) $192\pi \text{ cm}^3$

39. If it took a speed boat 9 hours to travel a distance of 1080 km, what was its average speed?

- (A) 12 km/h
- (B) 102 km/h
- (C) 120 km/h
- (D) 1200 km/h

40. The pie-chart shows the preference in drinks of a group of students.



If 12 students prefer chocolate, then the total number of students in the group is

- (A) 48
- (B) 72
- (C) 180
- (D) 360

41. The median of the numbers 1, 1, 5, 5, 6, 7, 7, 7, 8 is

- (A) 7
- (B) 6.5
- (C) 6
- (D) 5.4

42. If the mean of the four numbers 4, 8, x and 12 is 10, then x is

- (A) 4
- (B) 10
- (C) 12
- (D) 16

43. The highest weekly wage of a group of employees is \$105.40. If the range of the wages is \$27.50, how much does the lowest paid employee receive?

- (A) \$105.40
- (B) \$ 77.90
- (C) \$ 66.45
- (D) \$ 27.50

44. A bag contains 4 red balls and 5 blue balls. A ball is picked at random from the bag and is found to be red. It is not replaced. What is the probability that the next ball to be taken randomly from the bag will be blue?

- (A) $\frac{1}{5}$
- (B) $\frac{5}{9}$
- (C) $\frac{3}{5}$
- (D) $\frac{5}{8}$

Item 45 refers to the contents of the following table which shows the results of a survey of 100 households conducted by Form 5 students to determine the number of children in each household.

No. of children at home	0	1	2	3	4	5	6	7
Frequency	5	10	44	27	8	5	1	0

45. The probability that a home visited at random will contain exactly 4 children is

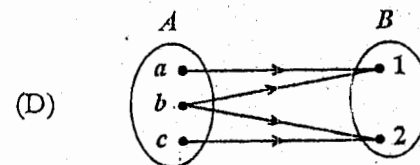
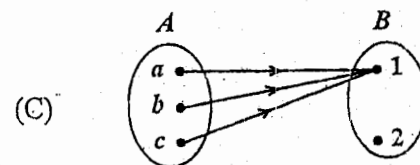
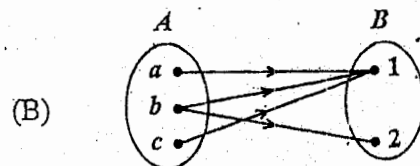
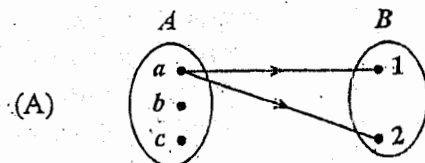
(A) $\frac{4}{100}$

(B) $\frac{8}{100}$

(C) $\frac{4}{28}$

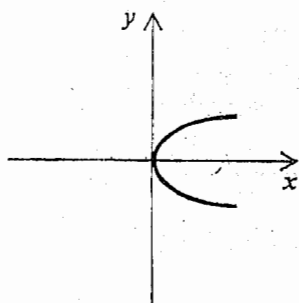
(D) $\frac{4}{8}$

46. Which of the following mappings from set A to set B is a function?

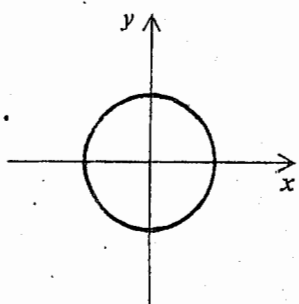


47. Which two graphs below represent functions?

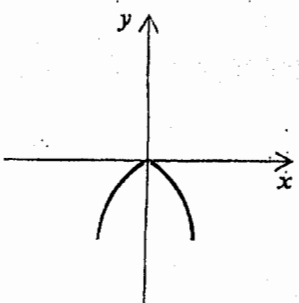
I



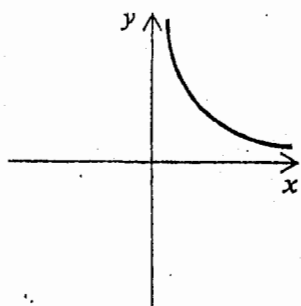
II



III



IV

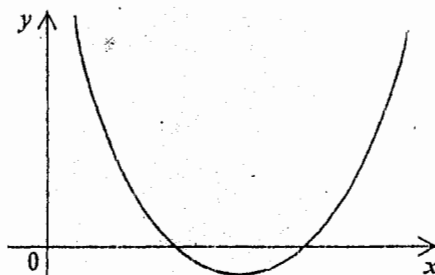


- (A) I and II
(B) I and III
(C) II and IV
(D) III and IV

48. If $h(x) = \frac{3x-2}{5}$ then $h(-6) =$

- (A) -4
(B) $-\frac{16}{5}$
(C) $\frac{16}{5}$
(D) 4

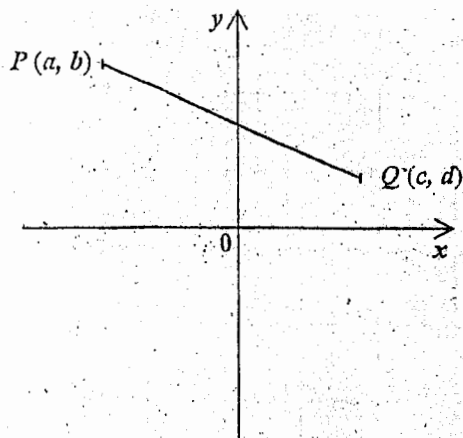
Item 49 refers to the following graph.



49. The diagram above shows a graph. If a , b and c are positive constants, the equation of the graph could be

- (A) $y = ax^2 + c$
(B) $y = c - ax^2$
(C) $y = ax^2 + bx + c$
(D) $y = c + bx - ax^2$

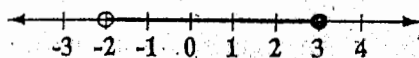
Item 50 refers to the following diagram.



50. The diagram above shows a line PQ . The gradient of the line PQ is given by

- (A) $\frac{b-d}{c-a}$
 (B) $\frac{c-a}{b-d}$
 (C) $\frac{a-c}{b-d}$
 (D) $\frac{b-d}{a-c}$

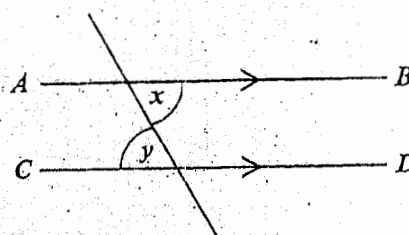
Item 51 refers to the diagram below.



51. The graph of the inequality in the diagram above is defined by

- (A) $-2 < x \leq 3$
 (B) $-2 \leq x < 3$
 (C) $-2 > x \geq 3$
 (D) $-2 \geq x > 3$

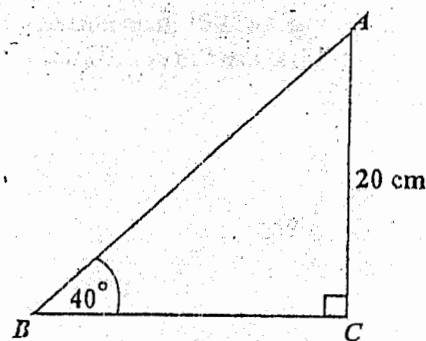
Item 52 refers to the following diagram.



52. In the figure above AB and CD are parallel. Which of the following BEST describes the relation between x and y ?

- (A) $x + y < 2x$
 (B) $x = y$
 (C) $x + y > 2x$
 (D) $x < y$

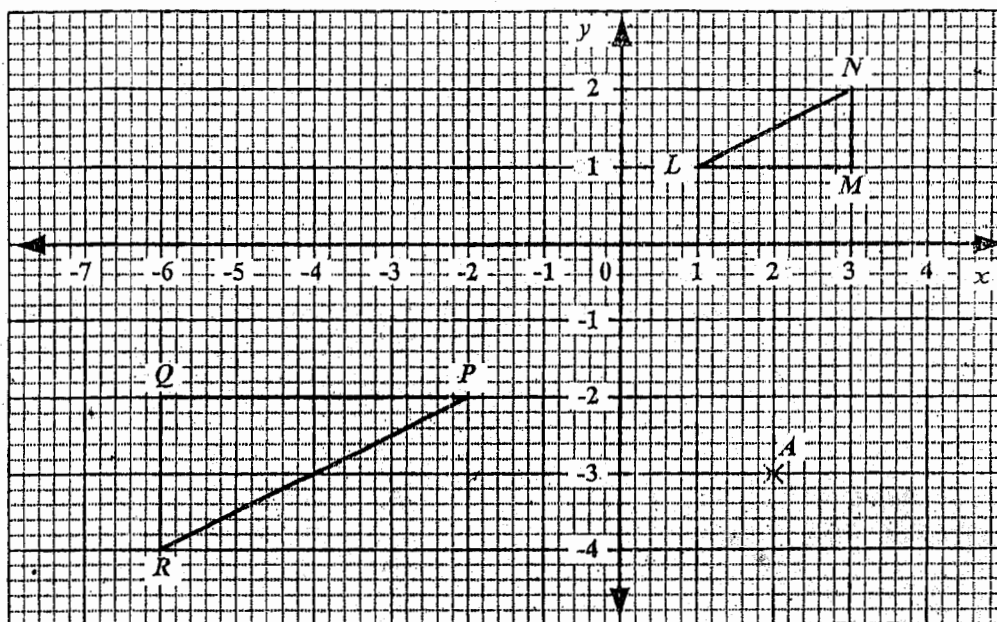
Item 53 refers to the following diagram.



53. The triangle ABC above is right-angled at C . $\angle ABC = 40^\circ$ and $AC = 20\text{cm}$. The length of BC , in cm, is

- (A) $20 \sin 40^\circ$
 (B) $\frac{20}{\sin 40^\circ}$
 (C) $20 \tan 40^\circ$
 (D) $\frac{20}{\tan 40^\circ}$

Item 54 - 55 refer to the following graph.



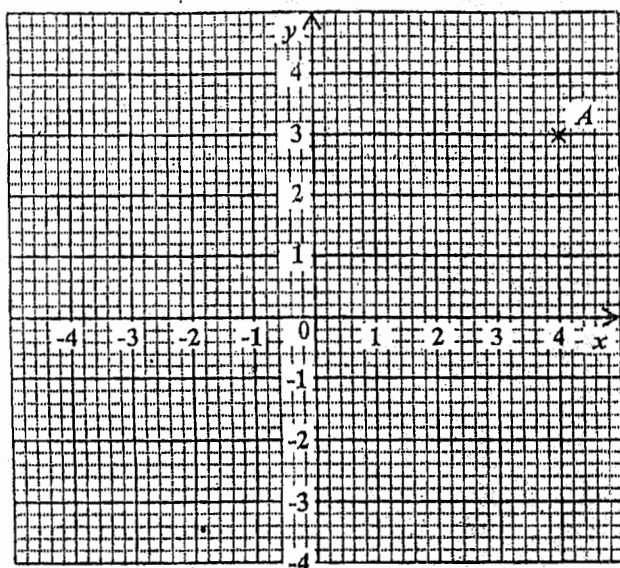
54. The point $A(2, -3)$ is rotated about the origin through an angle of 90° in an anticlockwise direction. What are the coordinates of the image of A ?

(A) $(3, 2)$
 (B) $(2, 3)$
 (C) $(-3, -2)$
 (D) $(-3, 2)$

55. The transformation that maps $\triangle LMN$ onto $\triangle PQR$ is

(A) a rotation through 180° about the origin
 (B) a rotation of 180° about $\left(\frac{-1}{2}, \frac{-1}{2}\right)$
 (C) an enlargement about $\left(\frac{1}{2}, \frac{-1}{2}\right)$ of scale factor 2
 (D) An enlargement about the origin of scale factor -2

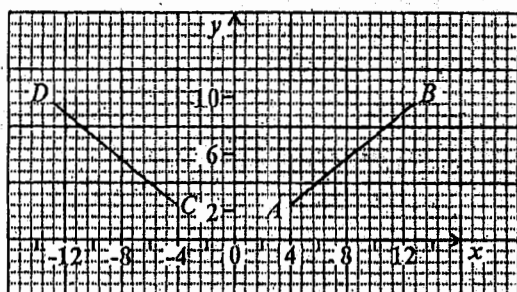
Item 56 refers to the following graph which shows the point A.



What are the co-ordinates of the image of A under reflection in the y-axis?

- (A) (-3, 4)
- (B) (3, -4)
- (C) (4, -3)
- (D) (-4, 3)

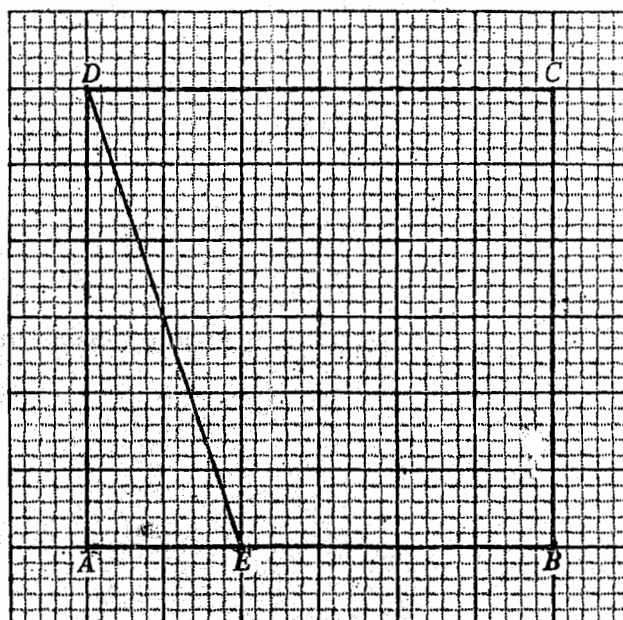
Item 57 refers to the following graph.



57. In the figure above, the line CD is the image of AB after

- (A) a rotation through 90° centre O
- (B) a reflection in the y-axis
- (C) a translation by vector $\begin{pmatrix} -4 \\ -8 \end{pmatrix}$
- (D) an enlargement of scale factor -1

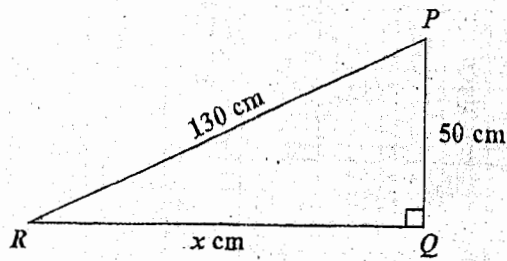
Item 58 refers to the following diagram.



58. How many triangles congruent to $\triangle ADE$ would be needed to cover the rectangle $ABCD$ entirely?

- (A) 2
- (B) 4
- (C) 6
- (D) 8

Item 59 refers to the following diagram.

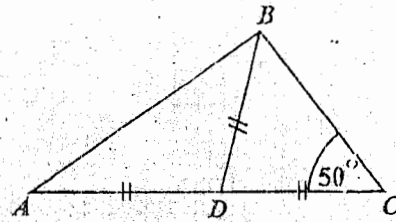


59. In the right-angled triangle above, not drawn to scale, $\hat{Q} = 90^\circ$, $PQ = 50\text{ cm}$, $PR = 130\text{ cm}$, and $RQ = x\text{ cm}$.

$\tan \hat{P}RQ =$

- (A) $\frac{50}{x}$
(B) $\frac{x}{50}$
(C) $\frac{50}{130}$
(D) $\frac{x}{130}$

Item 60 refers to the following diagram.



60. In the figure above, ABC is a triangle in which $AD = BD = CD$.

The angle ABC is

- (A) 90°
(B) 80°
(C) 50°
(D) 40°

IF YOU FINISH BEFORE TIME IS CALLED, CHECK YOUR WORK ON THIS TEST.